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THE CENSUS — all Boolean functions on 4 variables

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functions to examine: 65,536 inputs each: 16

Parseval check across all 65,536 functions: min 1.000000000 max 1.000000000

computing block sensitivity, decision-tree depth, certificate complexity ...

... 16,384 done

... 32,768 done

... 49,152 done

... complete

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[A] THE LANDSCAPE — exact distributions over all 65,536 functions

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sensitivity $s(f)$	0:	2	1:	8	2:	692	3:28,328	4:36,506
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block sens $bs(f)$	0:	2	1:	8	2:	668	3:28,352	4:36,506
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degree $deg(f)$	0:	2	1:	8	2:	212	3:12,648	4:52,666
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DT depth $D(f)$	0:	2	1:	8	2:	156	3: 8,240	4:57,130
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certificate $C(f)$	0:	2	1:	8	2:	668	3:28,352	4:36,506
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Shannon's counting argument, made exact:

functions of FULL degree 4: 52,666 of 65,536 = 80.36%

-> hardness is the overwhelming default. This is 'largeness' in

Razborov-Rudich, measured rather than assumed.

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[B] HUANG'S SENSITIVITY THEOREM (2019), verified on every function

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claim: $s(f) \geq \sqrt{\deg(f)}$ for every Boolean f

functions checked : 65,536

VIOLATIONS FOUND : 0

verdict : HOLDS UNIVERSALLY

slack $s - \sqrt{\deg}$: min 0.0000 mean 1.6000 max 2.2679

functions achieving minimum slack: 104

the standard chain $s \leq bs \leq C \leq D$, tested exhaustively:

$s \leq bs$: holds

$bs \leq C$: holds

$C \leq D$: holds

$\deg \leq D$: holds

$bs \leq \deg^2$: holds

largest s-vs-bs separation at n=4: 1
witness f = 0x1BD8 (s=2, bs=3, deg=2, D=3)

UNPLANNED CHECK - C(f) vs bs(f): identical for 65,536 of 65,536 functions
C(f) = bs(f) for EVERY function at n=4.
In general only $bs \leq C \leq bs^2$ is guaranteed; the gap needs
more room than 4 variables provide. n=4 is too small to see it.

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[C] HUANG'S PROOF OBJECT - built from scratch
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The proof puts +/- signs on hypercube edges and reads the eigenvalues.

$A_1 = [[0,1],[1,0]]$; $A_k = [[A_{k-1}, I], [I, -A_{k-1}]]$

A_4 is 16x16

$A^2 = 4 \cdot I$? max deviation 0.00e+00 -> YES

eigenvalues +2.0000 x8 -2.0000 x8

every $|\lambda| = 2.0000$? True

Cauchy interlacing then forces: any induced subgraph of the
16-vertex hypercube on more than 8 vertices has a vertex of
degree $\geq \sqrt{4} = 2.0000$. That single spectral fact is the
whole theorem. Two pages. A thirty-year open problem.

verifying the graph statement by brute force on random large subsets:
20,000 random subsets of size > 8
minimum max-degree observed: 2 (theorem requires ≥ 2.0000)
-> CONSISTENT

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[D] THE BARRIER, MEASURED
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spectral detector: weight on levels ≥ 3 exceeds a threshold.

swept every threshold; BEST is 0.07 at 82.09% accuracy.

correct 53,800 missed 304 false alarms 11,432

HONEST READING - the crude detector is mediocre, not perfect.
Guessing 'hard' for everything already scores 80.36%,
so it beats the trivial baseline by only 1.73 points.
A single spectral statistic is a weak proxy for real complexity.
Fourier lower bounds work because they use the WHOLE spectrum's
structure under restriction, not one summary number.

But both barrier conditions are met, and measurably:

CONSTRUCTIVE - one transform, 64 operations, timed in ms

LARGE - 80.36% of all functions are full degree

constructive + large = natural. At n=4 there is no cryptography to

break, so nothing stops it. That is precisely why the barrier is invisible at this scale and fatal at the next one.

plots written: census-landscape.png, census-huang-matrix.png

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